2026 Competitive Events Preparation Workshops
High School Coding

Problem:

The *Collatz Conjecture* is one of the most famous math problems. This problem takes any positive integer to start. If it is even then divide it by 2, and if it's odd then multiply it by 3 and add 1. If we keep doing this process over and over again, eventually, the result will be 1.

For example, starting with the number 10:

1. 10 is even: 10 / 2 = 5
2. 5 is odd: 5 x 3 + 1 = 16
3. 16 is even: 16 / 2 = 8
4. 8 is even: 8 / 2 = 4
5. 4 is even: 4 / 2 = 2
6. 2 is even: 2 / 2 = 1

For the number 10, it takes 6 steps to reach the number 1.

Your Task:

Create a program that takes an input of any positive integer, n, then outputs the number of steps, s, it takes for n to get to 1, following the Collatz Conjecture.

Example:

Input: 10

Output: 6

Input: 22

Output: 15

Input: 79

Output: 35